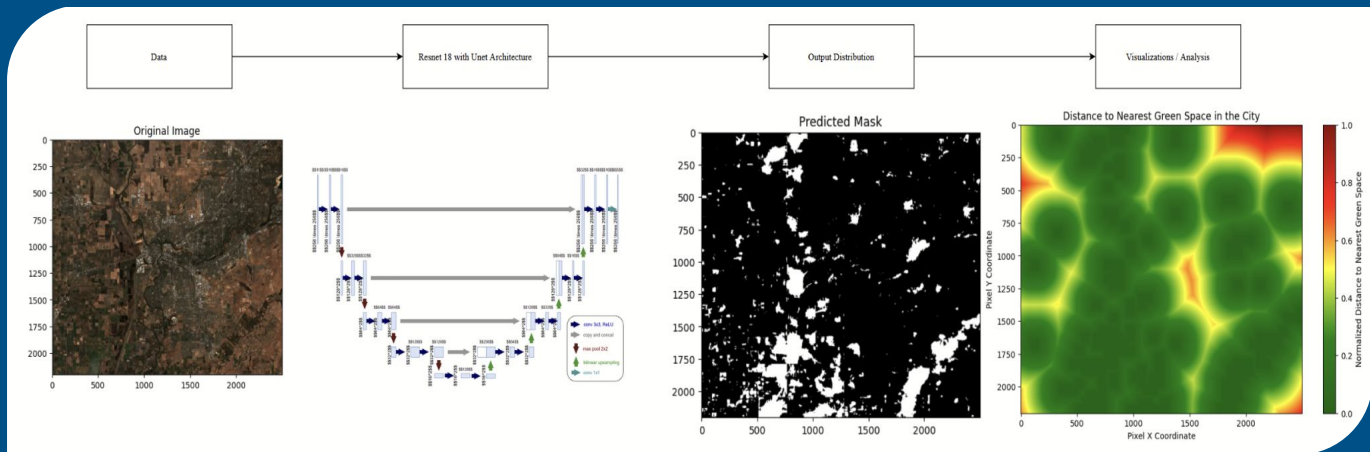


CNN-Based Urban Green Index

By Deven Biehler



1. **Defining the Problem**
2. **Methodology**
3. **Visual Analysis**
4. **Experiment Results**
5. **Drawing a Conclusion**

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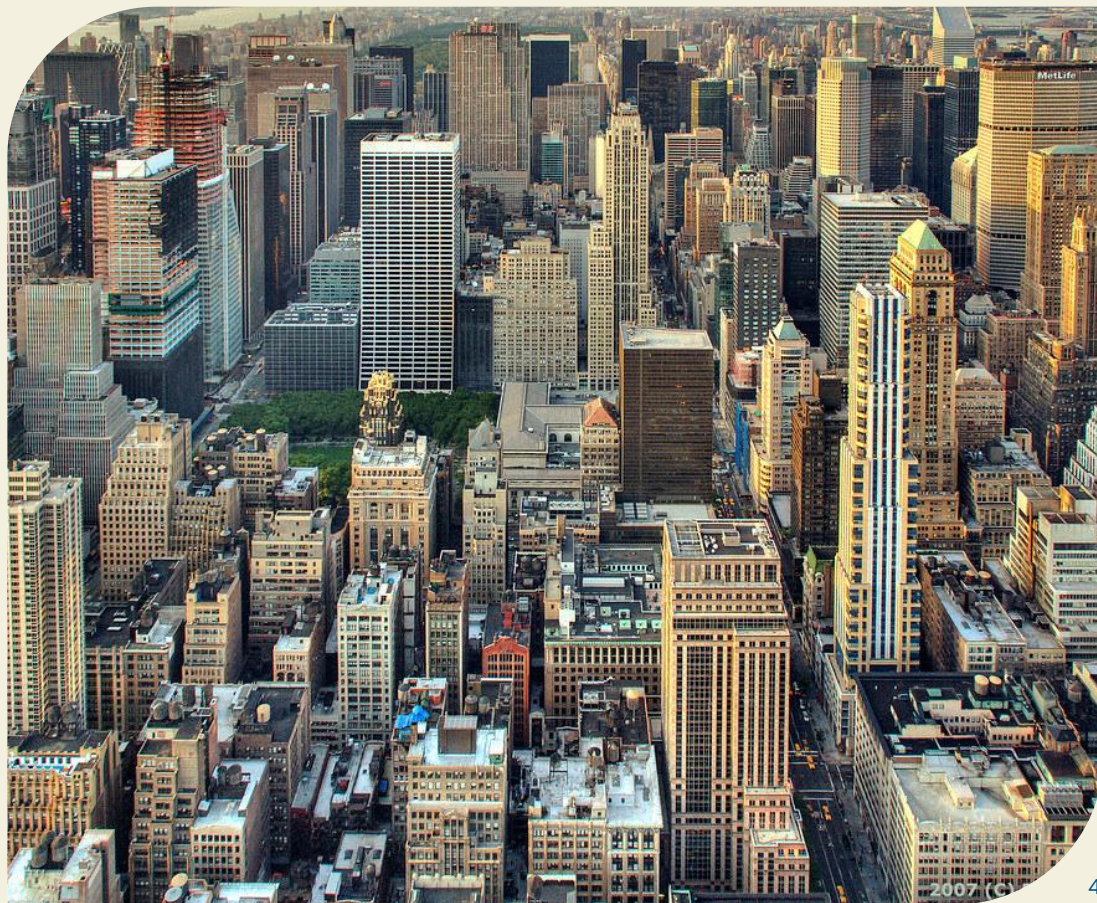
Problem

Green Index

- Distribution
- Score

Green Space

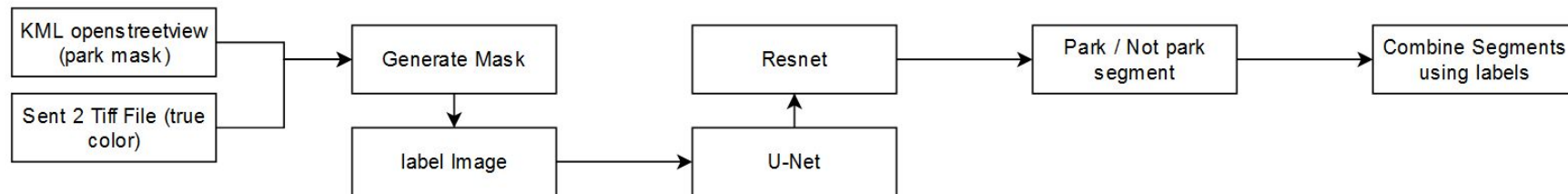
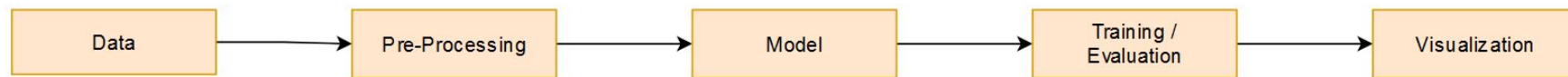
- Parks
- Gardens
- Golf course
- Etc.



"Cities have the capability of providing something for everybody, only because, and only when, they are created by everybody."

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Plan

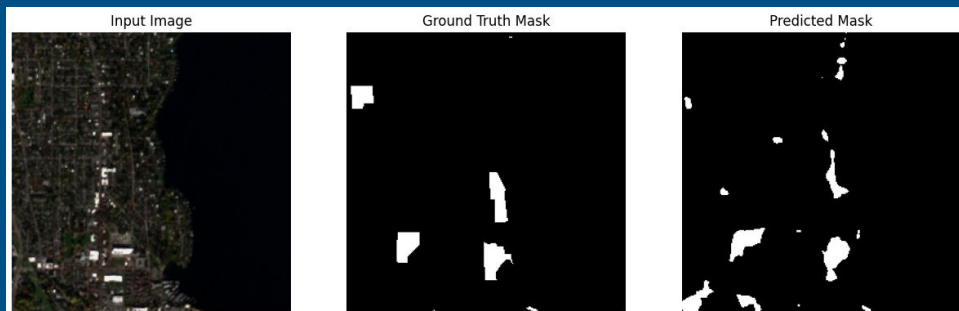
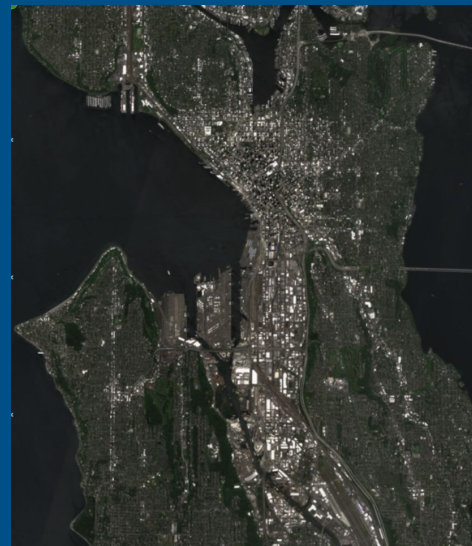


Data

Segmented city data from multiple cities.

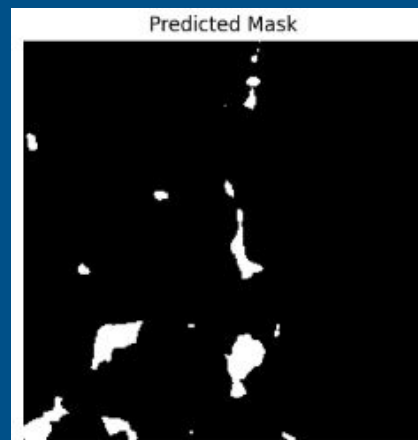
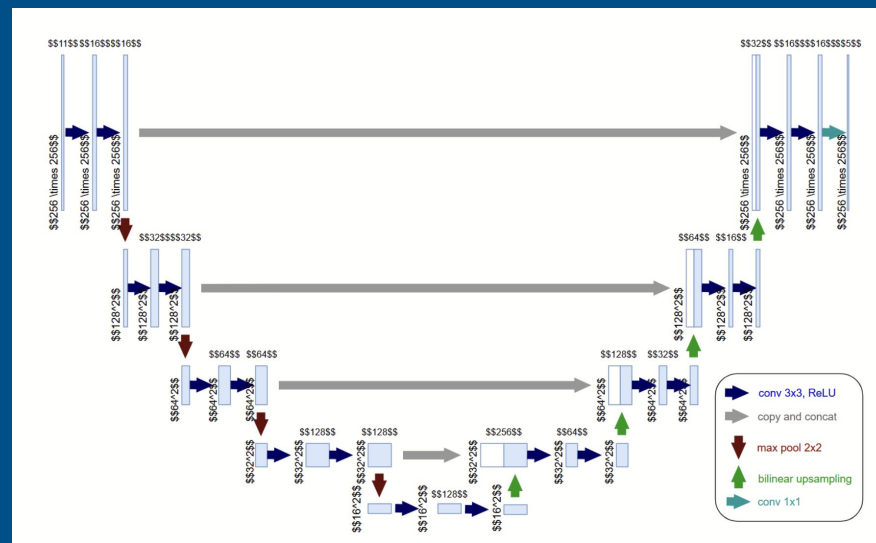
Satellite imagery from Sentinel 2 Satellite

Green space mask taken from OpenStreetMap



Model

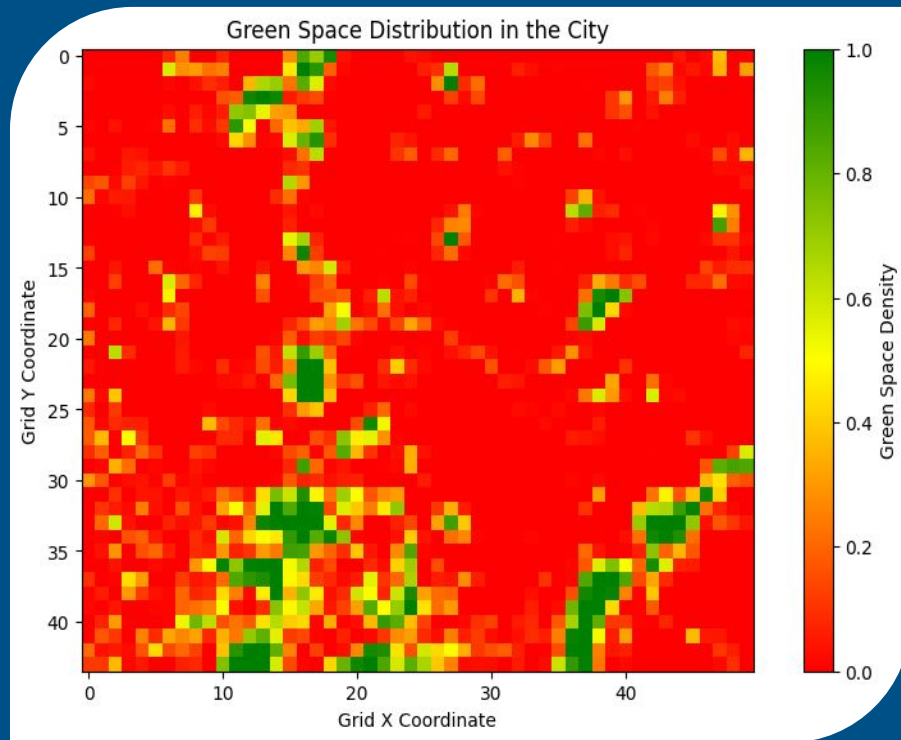
U-Net architecture with Resenet-34 backbone.



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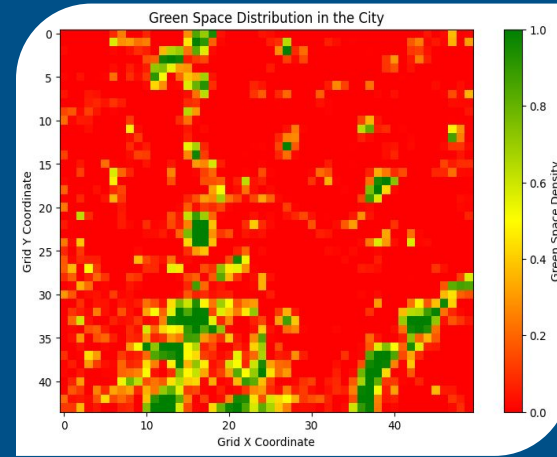
Green Density

Plot shows the density of green spaces throughout the city.



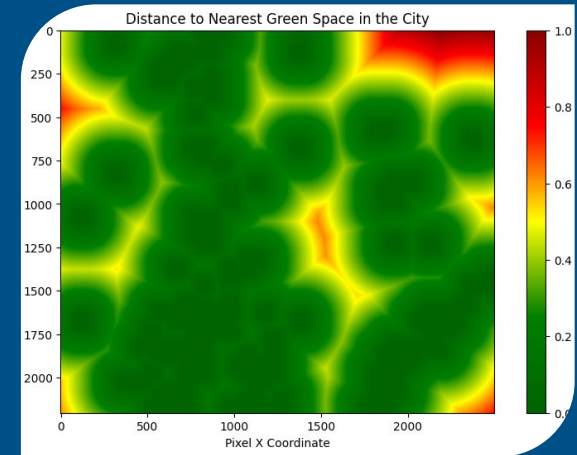
Green Density

Plot shows and interpretation of the density of green spaces throughout the city.



Labeled Green Spaces

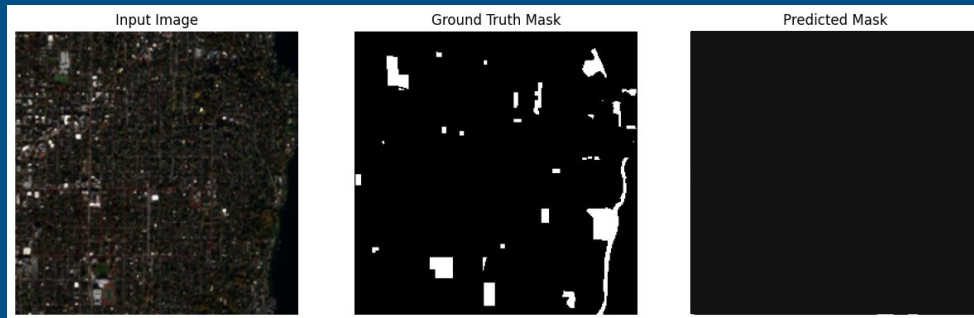
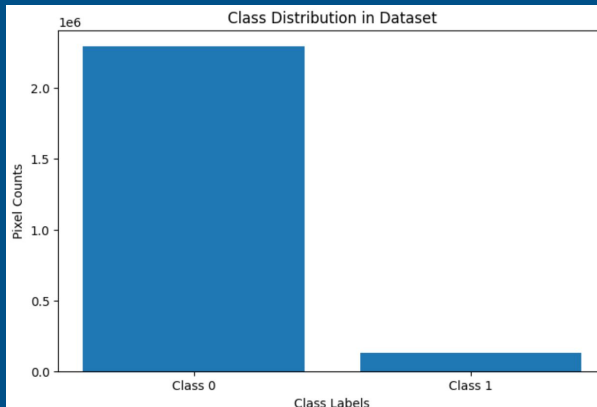
Green Space Density



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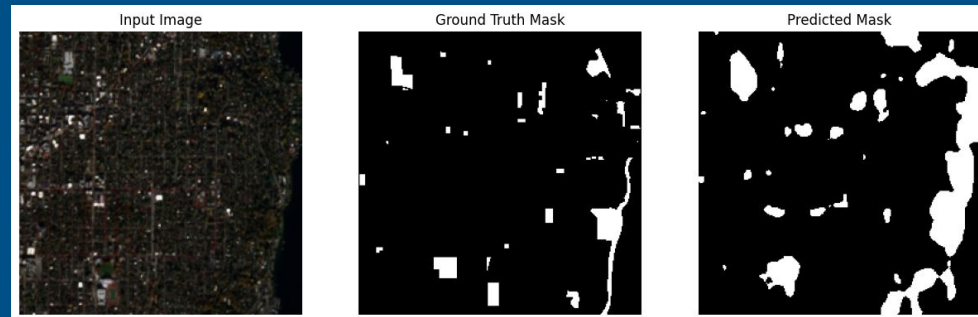
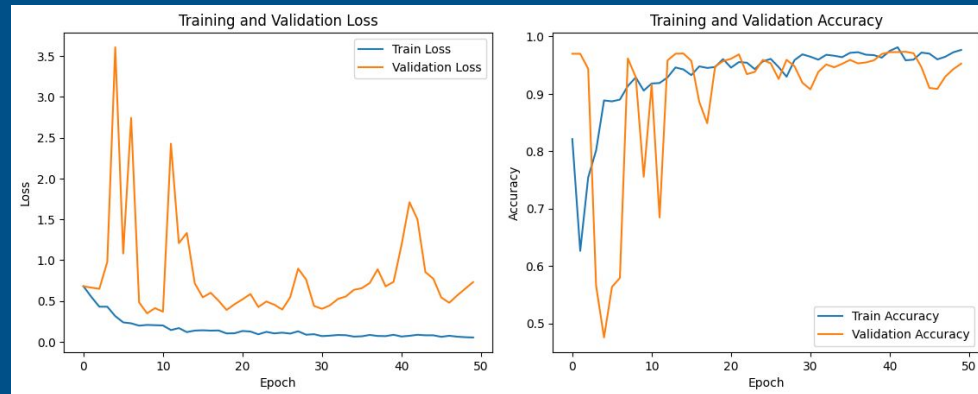
Large Class Imbalance

Due to the nature of my selected problem, there is a large class imbalance affecting the accuracy.



Accuracy / Loss

Good results on accuracy and loss after overcoming imbalance challenge.

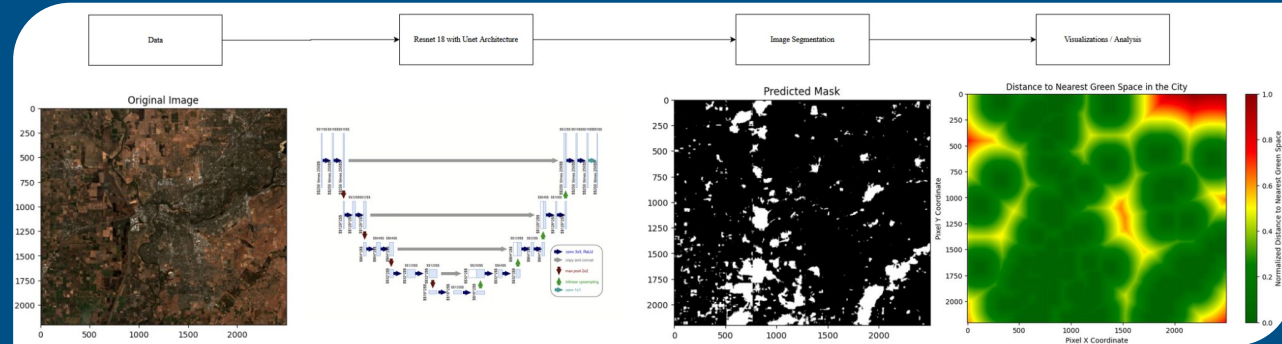


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Drawing a Conclusion

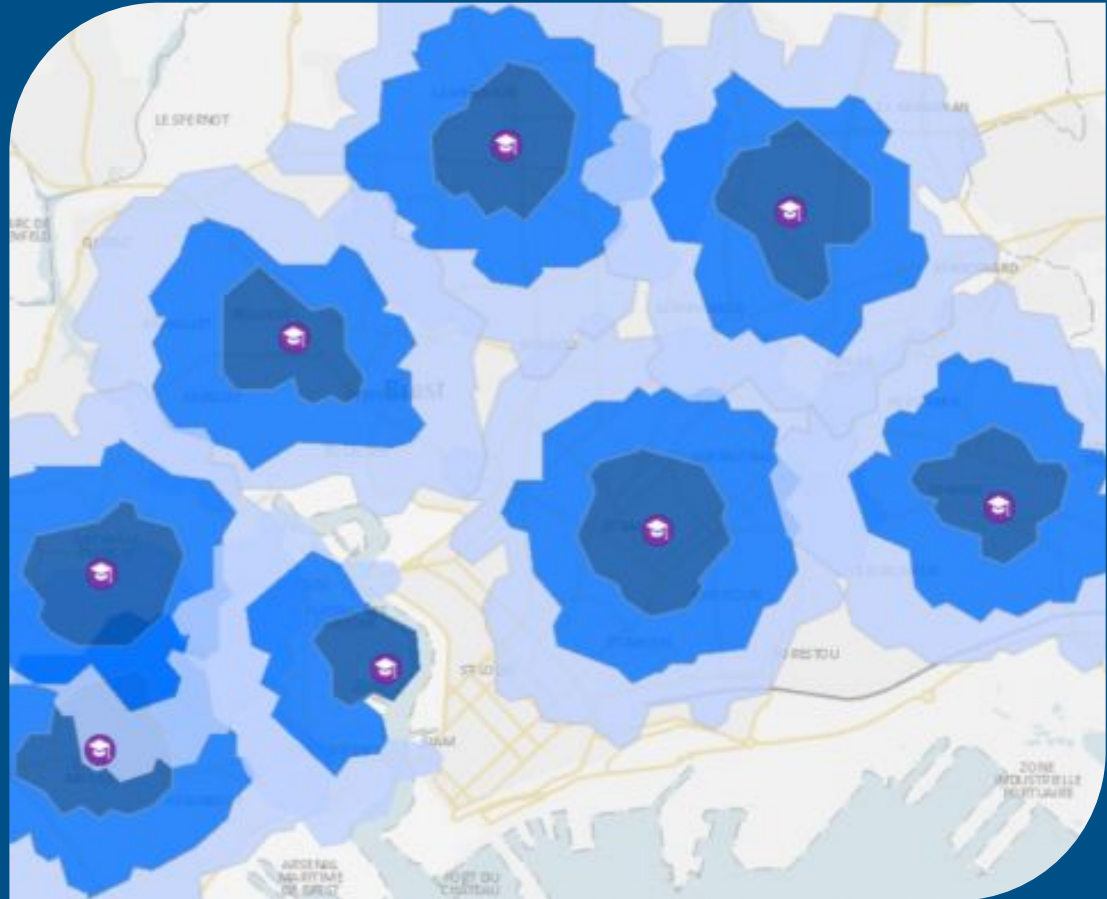
Implications:

- City Planning
- Home Buying/Selling
- Green Activists



Future Work

- Improving results through fine-tuning
- Improving visualizations through data agglomeration



Sentinel Hub

Eo browser,
<https://apps.sentinel-hub.com/eo-browser/>,
sinergise solutions

d.o.o., a planet labs company.

Source

Roberto E. Huerta, Fabiola D. Yépez, Diego F. Lozano-García, Víctor H. Guerra Cobán, Adrián L. Ferrino Fierro, Héctor de León Gómez, Ricardo A. Cavazos González, and Adriana Vargas-Martínez. Mapping urban green spaces at the metropolitan level using very high resolution satellite imagery and deep learning techniques for semantic segmentation. Remote Sensing, 13(11), 2021.

Overpass Turbo

Overpass turbo. <https://overpass-turbo.eu/>.

Source

A. Barbara Metzler, Ricky Nathvani, Viktoriia Sharmanska, Wenjia Bai, Emily Muller, Simon Moulds, Charles Agyei-Asabere, Dina Adjei-Boadi, Elvis Kyere-Gyeabour, Jacob Doku Tetteh, George Owusu, Samuel Agyei-Mensah, Jill Baumgartner, Brian E. Robinson, Raphael E. Arku, and Majid Ezzati. Phenotyping urban built and natural environments with high-resolution satellite images and unsupervised deep learning. Science of The Total Environment, 893:164794, 2023.

neighborhood green index

Kshama Gupta, Pramod Kumar, S.K. Pathan, and K.P. Sharma. Urban

neighborhood green index – a measure of green spaces in urban areas.

Landscape and Urban Planning, 105(3):325–335, 2012.

Source

OpenStreetMap contributors. Planet dump retrieved from <https://planet.osm.org>. <https://www.openstreetmap.org>, 2017.